

Lab 6: Multi-class Classification:

Logistic Regression vs. Neural Network

Objective:

- Explore and preprocess the Vinho Verde Wine Dataset.
- Implement and evaluate logistic regression and neural network for multi-class classification.
- Tune hyperparameters and compare the performance of both models.

Software Requirements:

- (1) Python
- (2) scikit-learn
- (3) pandas
- (4) numpy
- (5) matplotlib
- (6) torch

Dataset:

You may use the Vinho Verde Wine Dataset. This dataset contains information about red and white Vinho Verde wine samples from the north of Portugal, including physicochemical properties (such as acidity, alcohol content, etc.) and sensory quality scores. The target variable is the quality rating of the wine, which can be framed as a multi-class classification task.

Lab Instructions

1. Import necessary libraries.

2. Follow the exercises below to build and evaluate your multivariate linear regression model.

1) Data Loading and Exploration:

- Use Pandas to load the dataset.
- Display the first 5 rows of the dataset.
- Explore the dataset using methods like head(), describe(), info().
- Visualize distributions of key features using histograms or boxplots.
- Draw feature correlation heatmap.

2) Data Preprocessing:

- Check for missing values. If there are missing values, you need to handle them.
- Convert categorical variables into numeric format using One-Hot Encoding. (if any)
- Implement feature scaling.
- Split the data into training (80%) and test (20%) sets.

3) Model Building:

- Implement the multi-class logistic regression using the one-vs-rest strategy.
- Implement a neural network model with at least one hidden layer. (e.g. MLP)
- Train both models on the training data.
- Predict the labels for the test set and store them in variables.

4) Model Evaluation:

- Print out the classification report for both models.
- Generate and visualize confusion matrices for both models.

5) Hyperparameter Tuning:

5.1) For Logistic Regression:

- Use GridSearchCV to find the best hyperparameters for logistic regression.
- Tune parameters like 'C' (inverse regularization strength) and 'penalty' (type of regularization).
- Evaluate the model with the best parameters on the test set and compare the results with the untuned model.

5.2) For Neural Network:

- Use GridSearchCV to find the best hyperparameters for neural network.
- Tune parameters like the number of units in hidden layers, learning rate, etc.
- Train the neural network with the best hyperparameters and evaluate its performance.

6) Performance Comparison:

- Create a table or chart comparing the performances (e.g., accuracy, precision, recall, F1-score) of logistic regression and neural network.

Submission Requirement:

- 1) Complete sections 2 in a jupyter notebook file, name your file as “name_student number#.ipynb” and submit it through “Lab 6 Programming Assignment” on iSpace.
- 2) Please submit the executed code.

Deadline: 11:59pm, Mar.31, 2025, Late submission = 0; Plagiarism (cheating) = F